

Next, you combine these conditional probabilities or likelihoods with your prior probabilities to get the unconditional probability for DriveMed expanding, $P(\text{DriveMed expands})$, as follows:

$$\begin{aligned} &P(\text{DriveMed expands}) \\ &= P(\text{DriveMed expands} | \text{EPS exceeded consensus}) \times P(\text{EPS exceeded consensus}) \\ &+ P(\text{DriveMed expands} | \text{EPS met consensus}) \times P(\text{EPS met consensus}) \\ &+ P(\text{DriveMed expands} | \text{EPS fell short of consensus}) \times P(\text{EPS fell short of consensus}) \\ &= 0.75(0.45) + 0.20(0.30) + 0.05(0.25) = 0.41, \text{ or } 41\%. \end{aligned}$$

This is the total probability rule in action. Now you can answer your question by applying Bayes' formula, Equation 8:

$$\begin{aligned} &P(\text{EPS "exceeded" consensus} | \text{DriveMed "expands"}) \\ &= \frac{P(\text{DriveMed expands} | \text{EPS exceeded consensus})}{P(\text{DriveMed expands})} P(\text{EPS exceeded consensus}) \\ &= (0.75/0.41)(0.45) = 1.829268(0.45) \\ &= 0.823171 \end{aligned}$$

Before DriveMed's announcement, you thought the probability that DriveMed would beat consensus expectations was 45 percent. On the basis of your interpretation of the announcement, you update that probability to 82.3 percent. This updated probability is called your **posterior probability** because it reflects or comes after the new information.

The Bayes' calculation takes the prior probability, which was 45 percent, and multiplies it by a ratio—the first term on the right-hand side of the equal sign. The denominator of the ratio is the probability that DriveMed expands, as you view it without considering (conditioning on) anything else. Therefore, this probability is unconditional. The numerator is the probability that DriveMed expands, if last quarter's EPS actually exceeded the consensus estimate. This last probability is larger than unconditional probability in the denominator, so the ratio (1.83 roughly) is greater than 1. As a result, your updated or posterior probability is larger than your prior probability. Thus, the ratio reflects the impact of the new information on your prior beliefs.

EXAMPLE 4

Inferring Whether DriveMed's EPS Met Consensus EPS

You are still an investor in DriveMed stock. To review the givens, your prior probabilities are $P(\text{EPS exceeded consensus}) = 0.45$, $P(\text{EPS met consensus}) = 0.30$, and $P(\text{EPS fell short of consensus}) = 0.25$. You also have the following conditional probabilities:

$$P(\text{DriveMed expands} | \text{EPS exceeded consensus}) = 0.75$$

$$P(\text{DriveMed expands} | \text{EPS met consensus}) = 0.20$$

$$P(\text{DriveMed expands} | \text{EPS fell short of consensus}) = 0.05$$